Tideway submission notes

FinTechathon 2026 International Track, Topic E: SME Finance Copilot. This package is a synthetic research prototype. It connects to no real bank, lender, customer or supplier and moves no real money.

Team

The Last BrainCell

Bangladesh University of Engineering and Technology (BUET).

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Roles and team leader: awaiting the team's confirmation.

Reviewer quick start

Requires Python 3.11 or newer. No package installation, API key or paid service is required for the core sandbox. From the source directory:

```
python run.py
```

Open `http://127.0.0.1:8787`. Choose Imran for the accountant demo or Nadia for the owner demo. Demo PINs: Imran `1357`, Nadia `2468`, Guest `0000`. Use a fresh database for a repeatable run:

```
python run.py --db data/reviewer-fresh.sqlite3 --port 8788
python -m unittest discover -s tests -v
python -m tideway.selftest
```

The public handbook is at `/guide`. The model-status card distinguishes the deterministic default from optional local-compatible and Apple Silicon inference. Receipt extraction and TideLM are optional, with limitations in `docs/MODEL_CARD.md`.

Deliverable map

Requirement	Package location
Architecture, algorithm and security design	`docs/architecture.pdf`, `docs/TECHNICAL_DOCUMENTATION.md`
Presentation (10-minute maximum)	`slides/pitch.pdf`, `docs/PITCH_SCRIPT.md`
Demo video, within 5 minutes	Deferred by the team; recording is not included in this package
Source, setup and tests	`source/` in the ZIP; root README, run.py and tests in the working copy
Permission tiers and known risks	`docs/security-assessment.pdf`, `docs/SECURITY_SELF_ASSESSMENT.md`
Sandbox execution evidence	`evidence/results.csv`, `run-log.jsonl`, `verification-summary.json`, `screens/`
Validation and package integrity	`evidence/test-summary.json`, `package-manifest.json` and ZIP SHA-256
Model origin, licenses and limits	`docs/MODEL_CARD.md`, `THIRD_PARTY_NOTICES`, `licenses/`

Team-controlled final steps

The supplied roster is included. Confirm the team leader and actual roles, eligibility, registration acknowledgement, submission route and reviewer access. Keep the supplied public names/institution in tracked `submission-team.public.json`; roles and leader stay blank until confirmed. Private administrative facts belong in ignored `submission-team.json` and are never copied into the source archive. Follow any organiser requirement for truthful AI-assistance disclosure. No roles, nationality, registration or consent have been invented.

Published instructions checked on 15 October 2026

Registration and submission deadline: 20 October 2026, 23:59 Beijing time, which is 21:59 Bangladesh time. English is the submission language. The published rubric assigns 40% task completion, 30% security/compliance and 30% innovation/ interaction. Sandbox evidence is accepted for non-on-chain topics. Video is listed as optional but also named under execution evidence; the team has explicitly deferred the recording. Add it before final submission or obtain written organiser clarification.

- Competition rules
- IEEE track instructions
- CMU announcement

IEEE lists `fintechathon@ican-x.com`; CMU and the rules list `wangyr@pbcsf.tsinghua.edu.cn`. Use the team's acknowledgement to confirm the route before sending personal documents. This package contains no passports.

Honest limits

All data and counterparties are synthetic. Identity and keys are demo-grade. TidePay federation is a single-process simulation with illustrative masks. TideLM has a synthetic template-based benchmark, not real-world accuracy. Post-fix TideVision inference on Apple Silicon remains unverified. There is no completed

SME user study, live cloud-provider benchmark or external penetration test. Docker instructions are supplied, but a successful Docker build must only be claimed on a machine where it was actually run.

Contest upgrade evidence

Use `docs/PAYMENT_DELAY_RUNBOOK.md` for the live linked approval/invalidation/reapproval sequence. `evidence/upgrade-v1/` contains the corrected chronological prediction benchmark, distinct scenario workflows and screening/performance measurements, real-model availability report and delay receipt. Historical observations are retained under `evidence/history/pre-upgrade-v1/` and `evidence/ml/`. See `docs/CONTEST_UPGRADE_REPORT.md` for measured results and remaining verification limits.

The package source is tested and browser-checked at the same source digest; `python3 scripts/verify_package.py` checks archive hashes, extracted tests and a new loopback server with Python `-S`, no Git or developer configuration. This local preparation does not establish registration or submission readiness. The main site is dynamic and its full latest notices remain unverified.

Readiness follow-up

Historical `docs/READINESS_UPDATE.md` describes the four-step first-use guide and residual-risk warnings, contextual screening regressions, tracked cross-platform CI and Windows real-model smoke. The compatible Qwen3 backend matched 6/6 original and 12/12 additional authored smoke requests; raw failures and generations are preserved separately from historical TideLM adapter results. No model/runtime binaries are bundled. Follow `docs/SETUP_WINDOWS_MAC.md` to provision and start the optional model; the core needs neither weights nor an API key. CI is configured and locally checked; hosted CI has not run. Team-controlled administrative facts and video remain deferred.

Current evidence v3

Start with `docs/READINESS_CURRENT.md`. The exact shipped TidePay model is evaluated on 7,427 new synthetic invoices; mean absolute timing error is 8.5585 days versus 10.2296 for the history heuristic. The low-history limitation is reported separately. Historical audit-model scores do not transfer to shipped weights. Native macOS Qwen3 first handled 28/32 sealed requests; all four misses and source snapshots are preserved alongside later guard regressions.

Independent message screening detected only 2/24 unfamiliar attacks and flagged 2/24 benign messages. Payment authority therefore remains outside the model and screening. Above-target plans require each approver's server-checked, plan-bound risk acknowledgement. Mock financing includes timing, total repayment, unverified eligibility and affordability through repayment.

`python3 scripts/prepare_release.py` rebuilds and checks one frozen LF source tree, refreshes browser and PDF evidence, builds the local ZIP and verifies its fresh extraction. `--checks-only` runs core checks; Node 22+ and Chrome/Edge are required for the full command. No weight download, cloud provider, push or submission occurs. Real-user study status is pending with zero participants; the consent-aware kit does not invent feedback.